Week 3 BSM Chooser Option Validation

This report uses the paper's Table 2 inputs as the fixed model parameters and simulates a two-stage geometric Brownian motion path. The simulation is stochastic, so row-by-row values are not expected to match the paper exactly:

Formula

$$V_{chooser} = C(S, K, T) + P(S, K e^{-r(T-t)}, T-t)$$

$$S_t = S_0 * \exp((mu - 0.5 * sigma^{2)} * t + sigma * sqrt(t) * Z)$$

Replication Setup

- Random seed: 17170
- Price drift used in the GBM path: r - q = 0.0015 - 0.0233 = -0.0218
- Comparison rule: paper Table 3 rows are treated as a published reference, while the simulated rows come from the same Table 2 parameters under independent random draws

Parameters

- Spot price: 156.7
- Strike price: 150.0
- Risk-free rate: 0.0015
- Dividend yield: 0.0233
- Volatility: 0.282
- Decision time: 0.5
- Maturity: 1.0
- Source note: Week 3 paper table: JPM stock price 156.7, risk-free rate 0.15%, sigma 28.2%, strike 150, dividend yield 2.33%, decision time 0.5 years, maturity 1 year

Base Case

metric	value
spot_price	156.700000
strike_price	150
adjusted_strike_for_put_leg	149.887542
risk_free_rate	0.001500
dividend_yield	0.023300
volatility	0.282000
decision_time_years	0.500000
maturity_years	1

metric	value
time_to_decision_years	0.500000
call_leg_value	18.688997
put_leg_value	9.713217
chooser_value	28.402215
chooser_premium_vs_call	9.713217
chooser_premium_vs_put	18.688997
chooser_identity_gap	0

Spot Sensitivity

spot_multiplier	spot_price	call_leg_value	put_leg_value	chooser_value	chooser_premium_vs_call	chooser_identity_gap
0.850000	133.195000	7.763969	22.484730	30.248699	22.484730	0
0.950000	148.865000	14.470977	13.154096	27.625073	13.154096	0
1	156.700000	18.688997	9.713217	28.402215	9.713217	0
1.050000	164.535000	23.434825	7.017012	30.451837	7.017012	0
1.150000	180.205000	34.302356	3.449730	37.752086	3.449730	0

Validation Notes

- The chooser value is the call leg plus the adjusted put leg.
- The identity gap should be numerically zero, up to floating-point rounding.
- The chooser premium over the call leg is always the value of the added put leg.

Table 3 Simulation

The table below is generated from the Table 2 inputs. The exact path values depend on the seed and random draws, so they should be compared as a stochastic replication rather than a row-perfect match.

Simulation Summary

metric	value
rows_simulated	10
spot_price	156.700000
strike_price	150
risk_free_rate	0.001500
dividend_yield	0.023300
risk_neutral_drift_r_minus_q	-0.021800

metric	value
volatility_from_current_data	0.282000
random_seed	17170
decision_time_years	0.500000
maturity_years	1

Simulated Paths

row	z1	z2	simulated_st1	choice_call_put	simulated_st2	payoff
1	2.456421	-0.973346	247.986098	CALL	198.046618	48.046618
2	-0.189113	-0.149580	146.326778	PUT	137.721638	12.278362
3	0.831693	1.003948	179.360048	CALL	212.471007	62.471007
4	-0.526638	-0.204495	136.802515	PUT	127.355215	22.644785
5	-0.170727	0.145113	146.864221	PUT	146.593513	3.406487
6	0.849695	1.644397	180.005043	CALL	242.282243	92.282243
7	-0.819602	1.162799	129.039707	PUT	157.780688	0
8	1.350986	-1.014785	198.928276	CALL	157.560728	7.560728
9	1.558115	0.830742	207.316523	CALL	237.251103	87.251103
10	0.126236	-0.242662	155.823518	CALL	143.962835	0

Paper Reference Table

The next table shows the paper's published values only as a visual reference for comparison.

row	paper_st1	paper_choice	paper_st2	paper_payoff
1	118.330000	PUT	116.770000	33.230000
2	222.630000	CALL	192.890000	42.890000
3	186.530000	CALL	192.940000	42.940000
4	164.080000	CALL	148.770000	0
5	159.090000	CALL	116.780000	0
6	186.730000	CALL	128.120000	0
7	106.610000	PUT	90.520000	59.480000
8	163.060000	CALL	179.610000	29.610000
9	129.260000	PUT	144.820000	5.180000
10	115.410000	PUT	136.500000	13.500000

Aggregate Validation

This summary compares the simulated outputs against the paper at the distribution level. It includes the payoff mean, payoff standard deviation, payoff non-zero ratio, and the gap between the theoretical chooser value and the paper's average payoff.

metric	paper	simulated	difference
payoff_mean	22.683000	33.594133	10.911133
payoff_std	21.771920	36.188591	14.416670
payoff_nonzero_ratio	0.700000	0.800000	0.100000
theoretical_vs_paper_mean_gap	28.402215	22.683000	-5.719215
call_count	6	6	0
put_count	4	4	0

Interpretation: the paper's 10-row payoff sample is the reference distribution, while the simulated 10-row payoff sample comes from the same Table 2 parameters but different random draws. A non-zero gap is expected, and the statistics above show whether the two samples are in the same range.

Side-by-Side Comparison: Prices

This split table keeps the price columns readable and shows the row-by-row differences directly.

row	paper_st1	simulated_st1	st1_diff	paper_st2	simulated_st2	st2_diff
1	118.330000	247.986098	129.656098	116.770000	198.046618	81.276618
2	222.630000	146.326778	-76.303222	192.890000	137.721638	-55.168362
3	186.530000	179.360048	-7.169952	192.940000	212.471007	19.531007
4	164.080000	136.802515	-27.277485	148.770000	127.355215	-21.414785
5	159.090000	146.864221	-12.225779	116.780000	146.593513	29.813513
6	186.730000	180.005043	-6.724957	128.120000	242.282243	114.162243
7	106.610000	129.039707	22.429707	90.520000	157.780688	67.260688
8	163.060000	198.928276	35.868276	179.610000	157.560728	-22.049272
9	129.260000	207.316523	78.056523	144.820000	237.251103	92.431103
10	115.410000	155.823518	40.413518	136.500000	143.962835	7.462835

Side-by-Side Comparison: Choice and Payoff

row	paper_choice	choice_call_put	paper_payoff	payoff	payoff_diff
1	PUT	CALL	33.230000	48.046618	14.816618
2	CALL	PUT	42.890000	12.278362	-30.611638

row	paper_choice	choice_call_put	paper_payoff	payoff	payoff_diff
3	CALL	CALL	42.940000	62.471007	19.531007
4	CALL	PUT	0	22.644785	22.644785
5	CALL	PUT	0	3.406487	3.406487
6	CALL	CALL	0	92.282243	92.282243
7	PUT	PUT	59.480000	0	-59.480000
8	CALL	CALL	29.610000	7.560728	-22.049272
9	PUT	CALL	5.180000	87.251103	82.071103
10	PUT	CALL	13.500000	0	-13.500000

Validation Conclusion

- The implementation now clearly distinguishes model inputs, simulation assumptions, and the paper's reference outputs.
- Exact path-level equality is not expected because the simulated Table 3 uses random draws, but the aggregate comparison makes the deviation visible.
- The most likely reasons for differences versus the paper are the random seed and the paper's own simulation settings, which are not fully specified in the table excerpt.
- The report now states the seed, the GBM update rule, and the drift assumption so the run is reproducible.